

Course: B. Tech.

Branch: Civil Engineering

Semester: V

Subject Code & Name: BTHM505 Project Management

Max Marks: 60

Date: 10/07/2024

Duration: 3.00 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

- A) Explain Phases in Project Management (CO1) 6

- B) A project schedule has following characteristics (CO1) 6

Activity	A	B	C	D	E	F	G	H	I	J	K	L
	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Time (Days)	4	1	1	1	6	5	4	8	1	2	5	7

Construct a Network Diagram

Compute the earliest event time and latest event time.

Determine the critical path and total project duration

- C) Find the Total float and free float associated with the project in Q.1 (B) (CO1) 6

Q.2 Solve Any Two of the following.

- A) Explain briefly Direct Cost and Indirect Cost for construction project. (CO2) 6

- B) Consider the following table summarizing details for project.

Activity		A	B	C	D	E	F	G	H	I	J
Predecessors		-	-	-	A	B	C	C	E,F	D	H,G
Duration (week)	To	5	1	1	1	1	1	2	4	2	2
	T _m	6	3	4	2	2	5	2	4	5	2
	T _p	6	3	4	2	3	5	3	5	5	3

Construct the project network diagram.

Find the expected duration and variance of each activity.

Find the critical path and expected project completion of time. (CO3) 6

- C) What is the probability of completing the project on or before 22 weeks?
In Q.2. (B) (CO3) 6

Q.3 Solve Any Two of the following.

- A) Write short notes on Central limit Theorem. (CO3) 6

- B) Differentiate between resource leveling and resource smoothing. (CO2) 6

- C) Explain steps in Resource Updating with an example. (CO2) 6

Q.4 Solve Any Two of the following.

- A) Write short notes on Demand and Supply. (CO4) 6
- B) A company has following details: (CO4) 6
- Fixed Cost = Rs. 20,00,000/-
- Variable Cost per Unit = Rs.100
- Selling price per Unit = Rs.200/-
- Find
- (a) The break even sales quantity
- (b) The break even sales
- C) Write short notes on Inflation. (CO4) 6

Q.5 Solve Any Two of the following.

- A) Explain use of Computers in Project Management. (CO5) 6
- B) Explain in detail the concept of Quality Control. (CO5) 6
- C) Explain Total Quality Management (TQM) with special reference to Philosophy of Juran. (CO5) 6

***** End *****